INCH-POUND

MIL-W-22759/86
9 September 1994

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE/POLYIMIDE INSULATED, NORMAL WEIGHT, SILVER COATED COPPER CONDUCTOR, 200°C, 600-VOLTS

This specification sheet has been approved by all departments and agencies within the Department of Defense; however, it is not intended for use in Naval aircraft or Naval Air Systems applications.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-22759.

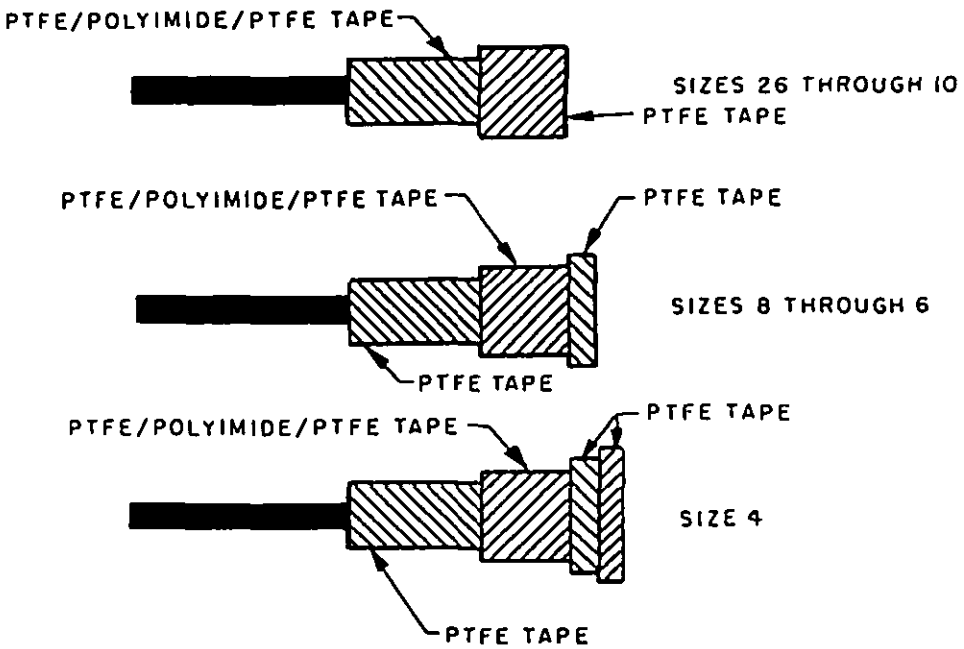


FIGURE 1. General configuration. 1/

1/ Small diameter stranded silver coated copper conductor (sizes 26 to 4).

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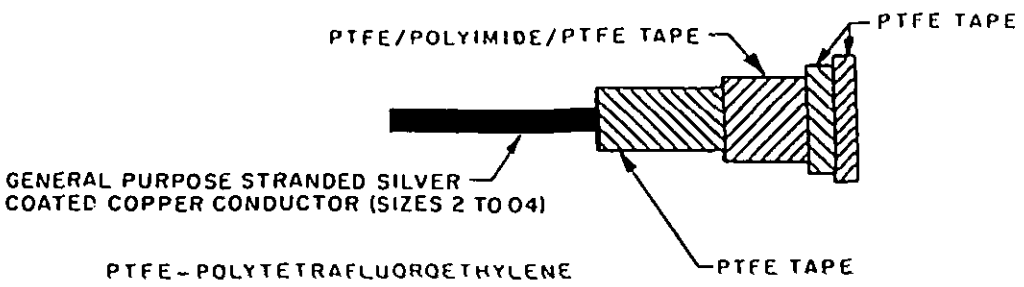


FIGURE 2. General configuration.

TABLE I. Construction details.

Part No. 1/	Wire Size	Conductor			Finished Wire			
		Stranding (Number of Strands X AWG Gauge of Strands)	Diameter (in.)		Resistance at 20°C (68°F) (ohms/ 1000 ft) (Max)	Diameter (in.)		Weight (lb/ 1000 ft) (Max)
			Min	Max		Min	Max	
M22759/ -26-*	26	19 x 38	0.018	0.020	38.4	0.033	0.037	1.6
M22759/ -24-*	24	19 x 36	0.023	0.024	24.3	0.038	0.042	2.2
M22759/ -22-*	22	19 x 34	0.029	0.030	15.1	0.043	0.047	3.1
M22759/ -20-*	20	19 x 32	0.037	0.038	9.19	0.051	0.055	4.7
M22759/ -18-*	18	19 x 30	0.046	0.048	5.79	0.064	0.068	7.4
M22759/ -16-*	16	19 x 29	0.052	0.054	4.52	0.068	0.073	8.8
M22759/ -14-*	14	19 x 27	0.065	0.068	2.88	0.081	0.086	13.2
M22759/ -12-*	12	37 x 28	0.084	0.087	1.90	0.100	0.105	20.2
M22759/ -10-*	10	37 x 26	0.106	0.110	1.19	0.122	0.127	32.2
M22759/ -8-*	8	133 x 29	0.158	0.166	0.658	0.180	0.188	58.0
M22759/ -6-*	6	133 x 27	0.198	0.208	0.418	0.219	0.229	90.0
M22759/ -4-*	4	133 x 25	0.250	0.263	0.264	0.276	0.288	143.0
M22759/ -2-*	2	665 x 30	0.320	0.340	0.170	0.344	0.364	221.0
M22759/ -1-*	1	817 x 30	0.360	0.380	0.139	0.388	0.408	288.0

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TABLE I. Construction details - Continued.

Part No. 1/	Wire Size	Conductor			Finished Wire			
		Stranding (Number of Strands X AWG Gauge of Strands)	Diameter (in.)		Resistance at 20°C (68°F) (ohms/ 1000 ft) (Max)	Diameter (in.)		Weight (lb/ 1000 ft) (Max)
			Min	Max		Min	Max	
M22759/ -01-*	0	1045 x 30	0.405	0.425	0.108	0.420	0.450	339.0
M22759/ -02-*	00	1330 x 30	0.440	0.475	0.085	0.475	0.505	424.0
M22759/ -03-*	000	1665 x 30	0.500	0.540	0.068	0.530	0.560	531.0
M22759/ -04-*	0000	2109 x 30	0.565	0.605	0.054	0.590	0.630	668.0

1/ Part number: The asterisks in the part number column of table I shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white with orange stripe - M22759/ - 20-93.

TABLE II. Wire insulation materials. 1/

Tape Code	Thickness (Nominal)	Material
1	0.002	0.0005(PTFE)/0.001(Polyimide)/0.0005(PTFE)
2	0.001	PTFE (Skived)
3	0.002	PTFE (Skived)
4	0.002	PTFE (Unsintered or presintered bondable)
5	0.0025	PTFE (Unsintered or presintered bondable)
6	0.003	PTFE (Unsintered or presintered bondable)

1/ Physical properties of PTFE tapes (skived and unsintered) shall be in accordance with MIL-W-22759 requirements.

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TABLE III. Physical properties of PTFE/Polyimide/PTFE tape.

Tensile Strength	12,000 lb/in^2 (average minimum)
Elongation	40 percent (average minimum)
Dielectric Strength	3,000 volts/mil (average minimum)

TABLE IV. Tape overlap requirements. 1/

Wire Size	Wrap 1		Wrap 2		Wrap 3		Wrap 4		Nominal Wall Thickness
	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	
26	1	50	4	50	-	-	-	-	.007
24	1	50	4	50	-	-	-	-	.007
22	1	50	4	50	-	-	-	-	.007
20	1	50	4	50	-	-	-	-	.007
18	1	50	4	50	-	-	-	-	.009
16	1	50	5	50	-	-	-	-	.008
14	1	50	5	50	-	-	-	-	.008
12	1	50	6	50	-	-	-	-	.009
10	1	50	6	50	-	-	-	-	.009
8	2	20	1	50	6	67	-	-	.014
6	2	20	1	50	6	67	-	-	.014
4	3	20	1	50	6	50	6	50	.017
2	3	20	1	50	6	50	6	50	.017

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TABLE IV. Tape overlap requirements - Continued. 1/

Wire Size	Wrap 1		Wrap 2		Wrap 3		Wrap 4		Nominal Wall Thickness
	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	
1	3	20	1	50	6	50	6	50	.017
0	3	20	1	50	6	50	6	50	.017
02	3	20	1	50	6	50	6	50	.017
03	3	20	1	50	6	50	6	50	.017
04	3	20	1	50	6	50	6	50	.017

1/ Wrap 1 is innermost tape which is in contact with the conductor. Wraps 2, 3, and 4 are progressively further away from the conductor core.

RATINGS:

Temperature rating: 200°C (392°F) maximum continuous conductor temperature.
Voltage rating: 600 Volts (rms) at sea level.

ADDITIONAL REQUIREMENTS:

Wet arc propagation resistance (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 3006. Measure the damage of the bundle along the axis. The wire is acceptable if the following criteria are met:

1. A minimum of 67 wires pass the dielectric test.
2. Three wires or less fail the dielectric test in any one bundle.
3. Actual damage to the wire is not more than 3 inches in any test bundle.

Dry arc propagation resistance (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 3007. Measure the damage of the bundle along the axis. The wire is acceptable if the following criteria are met:

1. A minimum of 67 wires pass the dielectric test.
2. Three wires or less fail the dielectric test in any test bundle.
3. Actual damage to the wire is not more than 3 inches in any test bundle.

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Blocking: 200°C ± 2°C (392°F ± 3.6°F).

Color: In accordance with MIL-STD-104, class 1; white preferred. Conformity of color to the limits of MIL-STD-104 shall not be required after oven exposure.

Color striping or banding durability: 125 cycles (250 strokes), 250 grams weight.

Conductor strand adhesion: Required.

Continuous lengths: Schedule B.

Dynamic cut-through (Test required for initial qualification only): Test size 20 only in accordance with ASTM D3032 section 22. Blade shall be the standard cutting blade except the cutting radius shall be .005 ±.001 inch. Minimum dynamic cut-through of 20 lbs at 23 ± 2°C (73°F ± 3.6°F) and 5 lbs at 200 ± 2°C (392°F ± 3.6°F).

Flammability: Test in accordance with MIL-STD-2223 method 1006 procedure A.
Requirements:

Duration of after-flame	3 seconds (max)
Flame travel	3.0 inches (max)
No flaming of tissue	

High-frequency spark test: (When used in lieu of Impulse Dielectric Test)
Test in accordance with MIL-STD-2223 method 3008, 5.7 kilovolts (rms).
Test 100 percent of the wire.

Humidity resistance: After humidity exposure, wire shall meet the requirements for initial insulation resistance.

Identification of product: Not required for size 26. Color code designator not required.

Identification durability: 125 cycles (250 strokes), 250 grams weight.

Immersion (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 1001 including additional fluids listed in Table V of this specification. Use mandrels and weights listed in Table VI for bend testing. Dielectric test, 2500 volts (rms), 60 Hz. For the turbine fuel immersion test of MIL-STD-2223, either Type JP-4 or MIL-T-83133 Type JP-8 (NATO Type F-34) may be used.

Impulse dielectric test: 8.0 kilovolts (peak). Test 100 percent of the wire.

Insulation resistance:
Sizes 26 through 10, 5000 megohms for 1000 feet (min).
Sizes 8 through 04, 3000 megohms for 1000 feet (min).

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Life cycle: 500 hours at 230°C ± 2°C (446°F ± 3.6°F). Dielectric test, 2500 volts (Rms), 60 Hz. Use mandrels coated with polytetrafluoroethylene such that the diameter of the mandrels, after coating, still conform to the required test mandrel diameters in Table VI. After oven exposure, layers shall not separate and/or tapes shall not lift along the insulation or at the ends.

Low temperature (cold bend): Use mandrels and weights specified in Table VI. Chamber temperature, -65°C ± 2°C (-85°F ± 3.6°F). Dielectric test, 2500 volts (rms), 60 Hz.

Shrinkage: Test at 230°C ± 2°C (446°F ± 3.6°F). Maximum change in measurement, 26-10 AWG 0.091 inch, 8-04 AWG 0.125 inch.

Smoke: 200°C ± 5°C (392°F ± 9°F); no visible smoke.

Solderability: Required.

Strippability: (Group II quality conformance test). Test sizes 26-14 only in accordance with ASTM D3032 section 27. The length of the insulation slugs shall be 0.25 inches. The maximum strip force shall be 4 lbs for sizes 20-26 and 6 lbs for sizes 18-14. No evidence of insulation left on the conductor when viewed with the naked eye.

Thermal Index: 200°C (392°F) minimum for 10,000 hours (Test required for initial qualification only). Test size 20 only in accordance with ASTM D3032, section 14.

Thermal shock resistance: Oven temperature, 200°C ± 2°C (392°F ± 3.6°F). Maximum change in measurement, 26-10 AWG 0.031 inch, 8-4 AWG 0.062 inch, 2-04 AWG 0.125 inch. No cracking.

Wrap (mandrel wrap): No cracking, no dielectric breakdown. Use mandrels specified in Table VI. No wrinkles shall be visible in the insulation at 3X magnification. Dielectric test, 2500 volts (rms) 60 Hz.

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TABLE V. Fluid table.

Test fluid	Test temperature (°C(°F))	Immersion time (hrs)
A. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, undiluted	48-50 (118-122)	20
B. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, diluted 60/40 (fluid/water) ratio	48-50 (118-122)	20
C. MIL-C-43616, Cleaning Compound, Aircraft Surface	48-50 (118-122)	20
D. ASTM D1153, Methyl Isobutyl Keyton (For use in Organic Coatings)	20-25 (68-77)	168
E. SAE-AS-1241, Fire Resistant Hydraulic Fluid for Aircraft	48-50 (118-122)	20
F. MIL-L-7808, Lubricating Oil, Aircraft Turbine Engine, Synthetic Base	118-121 (244-250)	30
G. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline Waterbase, undiluted	63-68 (145-154)	20
H. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline, Waterbase, diluted 25/75 (fluid/water) ratio	63-68 (145-154)	20
I. TT-S-735, Standard Test Fluids: Hydrocarbon, Type I	20-25 (68-77)	168
J. TT-S-735, Standard Test Fluids: Hydrocarbon, Type II	20-25 (68-77)	168
K. TT-S-735, Standard Test Fluids: Hydrocarbon, Type III	20-25 (68-77)	168

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TABLE V. Fluid table - Continued.

Test fluid	Test temperature (°C(°F))	Immersion time (hrs)
L. TT-S-735, Standard Test Fluids; Hydrocarbon, Type VII	20-25 (68-77)	168
M. Dielectric-coolant fluid, synthetic silicate ester base, Monsanto coolanol 25 or approved equivalent	20-25 (68-77)	168
N. MIL-G-3056, Gasoline, Automotive, Combat	20-25 (68-77)	168

TABLE VI. Test mandrel and test load requirements.

Wire Size (AWG)	Test Mandrel Diameter 1/ (in.)			Test Load 1/ (lb)	
	Cold Bend	Life Cycle/ Bend test	Wrap	Cold Bend	Life Cycle/ Bend test
26	1.00	0.375	0.125	3.00	0.50
24	1.00	0.500	0.125	3.00	0.75
22	1.00	0.500	0.125	4.00	1.00
20	1.00	0.500	0.125	4.00	1.50
18	1.50	0.750	0.250	5.00	2.00
16	1.50	1.00	0.250	5.00	2.00
14	2.00	1.00	0.375	5.00	3.00
12	2.00	1.50	0.375	5.00	3.00
10	3.00	2.00	0.375	6.00	3.00

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TABLE VI. Test mandrel and test load requirements - Continued.

Wire Size (AWG)	Test Mandrel Diameter 1/ (in.)			Test Load 1/ (lb)	
	Cold Bend	Life Cycle/ Bend test	Wrap	Cold Bend	Life Cycle/ Bend test
8	4.00	3.00	0.75	10.00	4.00
6	5.00	4.00	1.00	10.00	4.00
4	6.00	5.00	1.25	15.00	4.50
2	8.00	6.00	2.00	15.00	6.00
1	10.00	8.00	2.50	15.00	6.00
0	10.00	8.00	3.00	15.00	6.00
00	12.00	10.00	4.00	20.00	8.00
000	18.00	10.00	5.00	30.00	10.00
0000	18.00	10.00	6.00	30.00	10.00

1/ Tolerance shall be ± 3 percent of the given values.

Qualification of wire:

For qualification, a source is required to submit data on quality conformance tests and any finished wire tests as required by the qualification authorization letter. All other testing will be performed by the qualifying activity at the source's expense.

Due to the extended time period over which the thermal index test is performed, a source may become qualified under this specification sheet while this test is still in progress.

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Custodian:
Army - CR
Navy - AS
Air Force - 85
NASA - NA

Review Activities:
Army - AV, MI
Air Force. - 11, 19, 70, 99
DLA - ES, IS

Preparing Activity
Navy - AS

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